

15 / ENTITY
RESOLUTION

Entity Resolution Rules

Identifying items, suppliers and branches reliably

Purpose for the AI Assistant

How the assistant turns fuzzy user references ('bearing', 'Malik Iqbal', 'QCL') into exact database keys before any SQL runs. Wrong resolution is the biggest cause of wrong answers, so this step is mandatory.

Resolve first, query second

No department agent runs SQL until the entity is resolved to a key (item_code, canonical supplier, branch full-name). If resolution is ambiguous, the assistant asks one clarifying question instead of guessing.

1. Item resolution

Item names are not unique (thousands of 'Electrode' and 'Round Bar' variants). The assistant resolves in this strict priority order:

Priority	Signal	Action
1	Item Code given (e.g. 19981-60)	Exact match → done
2	Exact item name + specification	Match on item + specification
3	Item name only	May return many → narrow by category/branch
4	Category / group word ('bearing')	Filter item_sub_group → likely many

Worked example — “Show bearing stock”

Resolution path

1. 'bearing' matches item_sub_group='Bearings' (1,134 master items).
2. Too many to answer directly.
3. Respond with the aggregate AND offer to narrow:
'Bearings hold PKR X across N codes. Which bearing (code/spec) or which branch do you mean?'

Ambiguity rule

If step 3/4 yields more than a handful of matches, the assistant gives the category-level aggregate and asks one narrowing question — it never silently picks one item.

2. Supplier resolution

The same supplier appears with spelling and suffix variations across 194 names. The assistant normalises before matching.

- Lower-case, strip punctuation and legal suffixes ('(Pvt)', 'Ltd', '& Co', 'Enterprises').

- Match on the normalised core token(s); e.g. 'Malik Iqbal and Co...' → core 'malik iqbal'.
- Handle abbreviations via the synonym map (e.g. 'AJA' → 'A J A Enterprise').
- On multiple near-matches, present the top 2–3 canonical names and ask which one.
- Recommendation: back this with a supplier master table + canonical supplier_id (see Doc 10 gaps).

User says	Normalised	Resolves to
malik iqbal	malik iqbal	Malik Iqbal and Co.
AJA / A.J.A	aja	A J A Enterprise
nigar	nigar	Nigar Enterprises
umar nazir	umar a nazir	Umar A Nazir

3. Branch resolution

Purchase uses short codes; stock/issuance use full legal names. The assistant maps both directions so 'QCL' and 'Qadcast' are the same entity.

Code(s)	Canonical branch	Notes
QE, QEN	Qadri Engineering (Pvt) Ltd.	Machining & engineering
QCL	Qadcast (Pvt) Ltd.	Foundry / casting
QB2, QBL	Qadbros Engineering (Pvt) Ltd.	Fabrication & engineering
QB2-II, Unit-II	Qadri Brothers (Pvt.) Ltd. (Unit-II)	Unit-II operations
QE-II	Qadri Engineering – Unit II	Project entity (confirm w/ ERP)
IOL	(rare code in purchase data)	Confirm mapping w/ ERP

Confirm two mappings

QB2 vs QBL and the QE-II / IOL codes should be confirmed against the ERP before go-live (flagged in Doc 10). Until then the resolver treats them per the table above and states the assumption if used.

4. Resolution output contract

What the resolver returns to the agent

```
{
  "type": "item|supplier|branch",
  "status": "resolved|ambiguous|not_found",
  "key": "19981-60", // when resolved
  "candidates": [ ... ], // when ambiguous (top 2-3)
  "clarifying_question": "Which bearing spec did you mean?"
}
```